

# CATCH THE PHOTON · Facilitator Guide

A wave game with real phases · Grades 6–8

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| **Players** | 2 players |
| **Time** | about 10 minutes per game |
| **Big idea** | Waves can add up or cancel. Brightness is the wave's height squared. |

## Your job as the facilitator

You are the game guide. You do not need to be a quantum expert. Your job is to keep the game moving, read the rules out loud, and ask good questions. It is fine to say "Let's find out together."

## What the game is about

Each player has a detector, a spot where light can land. Playing cards sends waves to the detectors. Waves that are in step add up and make a detector **bright**. Opposite waves cancel and make it **dark**. At the end, one photon (one tiny bit of light) lands on a detector. The brighter detector has a better chance.

## What you need

- A deck of cards with only the A–10 cards (40 cards)
- The Wave Board printable
- One blue token (Player A) and one purple token (Player B)
- A cup and about 40 small chips or paper slips in blue and purple
- Optional: a computer with CatchThePhoton_Offline.html

## The four suits are four waves

| Suit | Wave | Moves the token |
|---|---|---|
| ♠ Spades | starts at a crest (top) | one square up |
| ♣ Clubs | starts at a trough (bottom) | one square down |
| ♥ Hearts | peaks a quarter later | one square right |
| ♦ Diamonds | peaks three quarters later | one square left |

Card numbers do not matter. ♠ and ♣ cancel. ♥ and ♦ cancel.

## Run the game, step by step

> **Say:** “Your goal is to make your detector bright and your opponent's detector dark. Then we catch the photon.”

1. Put both tokens on START (brightness 0, dark). Shuffle. Deal 3 cards to each player.
2. Both players secretly choose one card and lay it face-down next to a detector: **your own** (to make it brighter) or **your opponent's** (to cancel it).
3. Reveal together. Move that detector's token one square: ♠ up, ♣ down, ♥ right, ♦ left.
4. Draw back up to 3 cards. Play 8 rounds.

5 **Catch the photon.** Read the number under each token. Put that many chips of each player's color in the cup. Draw one chip without looking. That player wins.
6 If both numbers are 0, play one more round.

> **Edge of the board**
> A card cannot push a token off the board, so it cannot be played on that detector.

## Watch for these mistakes

- Thinking the card number matters. Only the suit counts.
- Moving the wrong token. The card moves the token of the detector it was played on.
- Adding the numbers on the board. The number is already the brightness; just read it.

## Try this together

Put 3 ♥ and 4 ♠ on one detector: 3 right and 4 up. The brightness is 25, because $3^2 + 4^2 = 5^2$. That is the Pythagorean theorem!

## Questions to ask

- When is it smart to play on your opponent's detector instead of your own?
- Why do ♠ and ♣ cancel?
- Two ♠ make a wave twice as tall. How much brighter is it? (4 times: $2^2 = 4$.)
- Why does a brighter detector have a better chance to catch the photon?

## The science words

| In the game | The real idea |
|---|---|
| Suit | Phase: where the wave's crest is |
| Waves adding up or canceling | Interference |
| Number on the square | Brightness = height$^2$ |
| Chips in the cup | The chance of finding the photon (the Born rule) |

Note: in real light, making one detector brighter makes the other dimmer. In this game both can grow. Real waves can also have any phase, not just four.

## Wrap up (say this)

> **Say:** “Quantum computers use this same trick. They set up waves so that the wrong answers cancel and the right answer gets bright.”